

Balanced & Unbalanced Forces

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Balanced forces cancel so motion does not change, while unbalanced forces produce a nonzero net force and change motion.
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Round 1 - Retrieve It

1. What is the net force when forces are balanced?

2. Can an object be moving while forces are balanced?

3. What is true about net force when forces are unbalanced?

4. What can an unbalanced force change?

Round 2 - Sketch It

Draw one balanced-force example and one unbalanced-force example. Use arrows with numbers and label the net force for each.

Balanced & Unbalanced Forces - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A book rests on a table. Gravity pulls down and the table pushes up equally. Balanced or unbalanced?

2. A car moves straight at constant speed. Are the overall forces necessarily unbalanced?

3. A soccer ball speeds up after being kicked. Balanced or unbalanced during the speed change?

Round 4 - Reason from Evidence

Explain why "balanced forces" does not mean "the object is not moving."

Balanced & Unbalanced Forces - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Balanced & Unbalanced Forces in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect balanced forces to Newton's first law. What stays the same when net force is zero?

E**Extend**

A skateboarder rolls at constant speed, then begins slowing. Describe how the force situation changed.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Balanced & Unbalanced Forces

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. 0 N.
2. Yes. It can move at constant speed in a straight line.
3. Net force is not zero.
4. An object's speed, direction, or both.

Sketch It - look for

Balanced example has equal opposing forces and 0 N net. Unbalanced example has unequal forces and a nonzero net force in the stronger direction.

Apply It

1. Balanced.
2. No. Constant velocity is consistent with balanced forces.
3. Unbalanced.

Reason from Evidence - look for

Balanced forces mean zero acceleration, not zero velocity. An object can remain at rest or keep moving at constant velocity.

ACE - Extend look-for

Initially net force was approximately zero. While slowing, an unbalanced force acts opposite the motion.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.